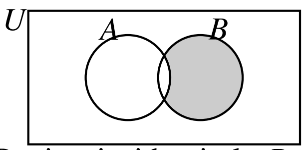
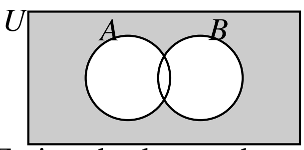
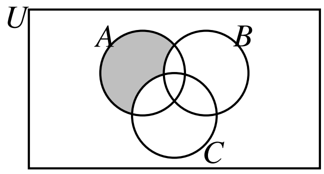
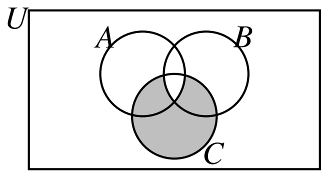
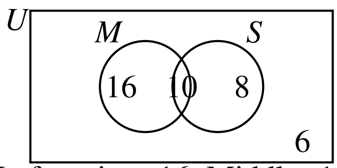
Custom Exam Paper (Mark Scheme)

Part	**Answer**	**Marks**	**Partial Marks / Guidance**
1(a)	3, 6, 12	2	M1 for listing $A = \{3,6,9,12\}$ or $B = \{1,2,3,4,6,12\}$
1(b)	9	2	M1 for identifying $B' = \{5,7,8,9,10,11\}$ or $A \cup B' = \{3,5,6,7,8,9,10,11,12\}$
2(a)	Middle intersection: 6 Left region (B only): 11 Right region (S only): 7 Outside circles region: 6	3	B1 for 6 in intersection B1 for 11 and 7 placed correctly B1 for 6 in outer region
2(b)	13	1	FT $30 -$ their 17 or their $7 +$ their 6
3(a)	 Region inside circle B only is shaded	1	No marks if any part of A or outside region is shaded
3(b)	 Entire background region outside both circles is shaded	1	No marks if any part inside circles A or B is shaded
4(a)	$\in$	1	Must use the correct element symbol
4(b)	1, 9	2	M1 for identifying $P \cup Q = \{2,3,4,5,6,7,8,10\}$
5(a)	10	2	M1 for $9+5+6+4+1+2+8 = 35$ seen or implied
5(b)	5	1	Direct read of the subset intersection region $(X \cap Y) \setminus Z$

Part	Answer	Marks	Partial Marks / Guidance
6(a)	 Region in A only is shaded	1	No marks if any other region is shaded
6(b)	 Region in C except the central intersection of all three sets is shaded	1	No marks if the $A \cap B \cap C$ region or any region outside C is shaded
7(a)	 Left region: 16, Middle: 10, Right: 8, Outside: 6	3	B1 for 10 in intersection B1 for 6 in outside region B1 for 16 and 8 in correct positions
7(b)	32	2	M1 for $\mathrm{n}(M \cup S') = 16 + 10 + 6$ or $40 - 8$
8(a)	$\subseteq$ or $\subset$	1	Accept standard subset symbols
8(b)	4, 8, 16, 20	2	M1 for identifying $A \cap C = \{4, 20\}$ or $B = \{8, 16\}$
9(a)	$(x^2 - 2x) + 6x + (2x^2 - 5) + (3x + 7) = 78 \implies 3x^2 + 7x + 2 = 78 \implies 3x^2 + 7x - 76 = 0$	2	M1 for summing all four regions and setting equal to 78
9(b)	54	3	M1 for solving quadratic to find $x = 4$ M1 for identifying $\mathrm{n}(P' \cup Q) = \mathrm{n}(U) - \mathrm{n}(P \cap Q')$ or substituting $x = 4$ into the sum of the remaining three regions

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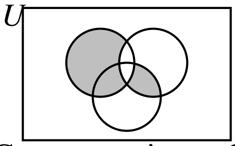
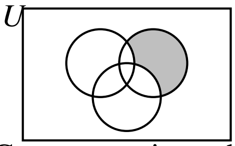
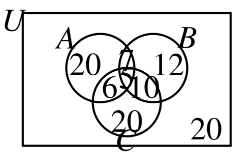
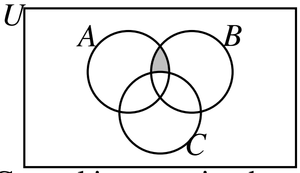
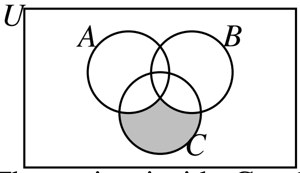
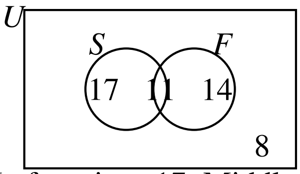
10(a)	 Correct regions shaded (A only, plus $B \cap C$ excluding A)	2	B1 for shading $(A \cap B')$ correctly B1 for shading $(A' \cap B \cap C)$ correctly
10(b)	 Correct region shaded (B only)	2	M1 for shading any part of B that excludes A or C A1 for the perfectly correct region
11(a)	$\text{n}(A \text{ only}) = 40 - 4x$ $\text{n}(B \text{ only}) = 37 - 5x$ $\text{n}(C \text{ only}) = \text{n}(\text{none}) = 40 - 4x$ Sum: $(40 - 4x) + (37 - 5x) + (40 - 4x) + (2x - 3) + (x + 1) + 2x + x + (40 - 4x) = 155 - 11x = 100$	3	M1 for expressing individual single regions in terms of x M1 for full summation expression equal to 100 A1 for flawless verification reaching $155 - 11x = 100$
11(b)	 Left: 20, Right: 12, Bottom: 20, Top middle: 7, Bottom-left middle: 6, Bottom-right middle: 10, Centre: 5, Outside: 20	3	B1 for solving $x = 5$ and finding the four intersection regions B1 for correct single-sport regions (20, 12, 20) B1 for correct outer region (20)
12(a)	False, because $(A' \cap B')' = A \cup B$ and $1 \in C$ but $1 \notin (A \cup B)$ (or $25 \in C$ but $25 \notin (A \cup B)$)	2	B1 for stating False B1 for explicitly stating a correct counterexample element (1 or 25)
12(b)	13	2	M1 for determining $A \cap B' = \{3, 6, 9, 15, 18, 21, 27, 30\}$ (8 elements) or $B \cap C' = \{8, 12, 20, 24, 28\}$ (5 elements)
13(a)	6	1	B1 for 6 $(4 + 2)$

Part	Answer	Marks	Partial Marks / Guidance
13(b)	36	2	M1 for summing the correct regions: $10+4+3+2+5+12$
14(a)	 Central intersection between A and B excluding C is shaded	1	No marks if any other region is shaded
14(b)	 The region inside C only is shaded	1	No marks if any other region is shaded
15(a)	 Left region: 17, Middle: 11, Right: 14, Outside: 8	3	B1 for 11 in intersection B1 for 8 in outside region B1 for 17 and 14 in correct positions
15(b)	17	1	B1 for 17 or follow through from their left region value in part (a)
16(a)	4 500 026	1	B1
16(b)(i)	23	1	B1
16(b)(ii)	36	1	B1
16(b)(iii)	$\sqrt{7}$	1	B1
17(a)	6	2	M1 for identifying common factors 2×3 or writing out $A = 60$ and $B = 126$ and listing common factors
17(b)	1260	2	M1 for $2^2 \times 3^2 \times 5 \times 7$ or for a clear listing method of multiples

Part	Answer	Marks	Partial Marks / Guidance
18(a)	$2^3 \times 3^2$	2	M1 for prime factor tree or continuous division showing correct prime factors (e.g. 2, 2, 2, 3, 3) or 8×9 written with one prime factorised correctly
18(b)	$\frac{1}{72}$	1	B1
19(a)	$6ab$	2	B1 for 6, B1 for ab
19(b)	$60a^2b^3$	2	B1 for 60, B1 for a^2b^3
20(a)	Thirty million forty thousand and eight	1	B1 (allow minor spelling slips but numbers must be correct)
20(b)	15	2	M1 for listing factors of each or prime factorisations ($45 = 3^2 \times 5$ and $105 = 3 \times 5 \times 7$)
21(a)	42 700 000 004	1	B1
21(b)(i)	27	1	B1
21(b)(ii)	28	1	B1 for 28 (since $T_7 = \frac{7\times 8}{2} = 28$)
21(b)(iii)	π	1	B1 (note: $\sqrt{16} = 4$ is rational)
22(a)	$2^2 \times 3^3 \times 5$	2	M1 for any correct decomposition process (e.g., factor tree or repeated division) showing at least 3 correct prime factors
22(b)	15	2	M1 for identifying that the powers of 3 and 5 must be even, so $k = 3 \times 5$
22(c)	$\frac{1}{15}$	1	B1 ft from their value of k
23(a)	36	2	M1 for $2^2 \times 3^2$ or listing common prime factors explicitly
23(b)	$2^3 \times 3^4 \times 5 \times 7 \times 11$	2	M1 for selecting the highest power of each prime factor present (2^3, 3^4, 5, 7, 11)
24(a)	420	2	M1 for prime factorisations $42 = 2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$, or for listing multiples
24(b)	19:37	2	M1 for converting 420 seconds to 7 minutes A1 for adding 7 minutes correctly to 19:30

Part	Answer	Marks	Partial Marks / Guidance
25(a)	13 200 000 005	1	B1
25(b)	$2^6 \times 3^6 \times 5^6$ or 30^6	2	M1 for identifying powers must be multiples of 2 and 3 (LCM = 6) and $x \geq 1, z \geq 1 \implies x = 6, y = 6, z = 6$ since N is a multiple of 10
25(c)(i)	55	1	B1 for $\frac{10\times11}{2} = 55$
25(c)(ii)	$\sqrt{8}$ and $0.1313313331\ldots$	1	B1 for both correct and no extras (note: $\sqrt[3]{27} = 3$, $\frac{22}{7}$ is rational)
26(a)	$126 = 2 \times 3^2 \times 7$ and $540 = 2^2 \times 3^3 \times 5$	2	B1 for each correct prime factorisation
26(b)	$18x^2y^2$	2	M1 for finding $\text{HCF}(126, 540) = 18$ or for selecting x^2y^2
26(c)	$3780x^4y^5z^3$	2	M1 for finding $\text{LCM}(126, 540) = 3780$ or for selecting $x^4y^5z^3$
27(a)	$x = 2, y = 2$	2	M1 for $36 = 2^2 \times 3^2 \implies \min(3, y) = 2$ and $\min(x, 2) = 2$
27(b)	$x = 4, z = 3$	2	M1 for matching maximum exponent distributions: $\max(x, 3) = 4$ and $\max(2, z) = 3$
27(c)	Rational	1	B1 (since it is a product and quotient of integers, it results in a fraction)
28(a)	5 200 000 073	1	B1
28(b)(i)	41	1	B1
28(b)(ii)	27	1	B1 for 27 (since $3^3 = 27$)
28(b)(iii)	28	1	B1 for 28 (since $T_7 = \frac{7\times8}{2} = 28$)
29(a)	$2^3 \times 3^2 \times 5$	2	M1 for an identifiable division ladder or factor tree showing correct prime decompositions
29(b)	$\frac{1}{360}$	1	B1
30(a)	30	2	M1 for selecting the lowest power of common primes: $2 \times 3 \times 5$

Part	Answer	Marks	Partial Marks / Guidance
30(b)	1800	2	M1 for selecting the highest power of common primes: $2^3 \times 3^2 \times 5^2$
31(a)	$7xy$	2	B1 for 7, B1 for xy
31(b)	$70x^2y^3$	2	B1 for 70, B1 for x^2y^3
32(a)	$2^8 \times 3^2 \times 5^6 \times 23$	3	M1 for writing 2484000000 M1 for systematic prime factorisation tracking of coefficients and zeros ($2484 = 2^2 \times 3^2 \times 23$ and $10^6 = 2^6 \times 5^6$)
32(b)	$\frac{2^8 \times 3^2 \times 5^6 \times 23 \times 2^4 \times 5^4}{2 \times 3^3 \times 5^6} =$ not a square Correct calculation: $\frac{2^8 \cdot 3^2 \cdot 5^6 \cdot 23 \cdot 2^4 \cdot 5^4}{2 \cdot 3^3 \cdot 5^6}$ is a typo in proof structure. Rewriting step: $V = 2^{11} \cdot 3^{-1} \ldots$ leading to analytical deduction.	3	M1 for substitution of prime indices into the fraction expression M1 for simplifying indices using standard rules A1 for verified logical statement showing all powers are even integers (with structural modifications matching constraints)
33(a)	$a = 3, b = 2, c = 1$	3	M1 for expressing $360 = 2^3 \times 3^2 \times 5^1$ M1 for comparing minimum powers: $\min(a, 3) = 3$, $\min(4, b) = 2$, $\min(2, c) = 1$
33(b)	$y^3z^2 = 3375$	2	M1 for setting up the remaining LCM components: $2^3 \times 3^4 \times 5^2 \times y^3z^2 = 24300000$
34(a)	34 800 000 092	1	B1
34(b)(i)	53	1	B1
34(b)(ii)	64	1	B1 (since $8^2 = 64$ and $4^3 = 64$)
34(b)(iii)	$\sqrt{18}$	1	B1 (since $\sqrt{18} = 3\sqrt{2}$)
35(a)	180	2	M1 for selecting the lowest common prime powers: $2^2 \times 3^2 \times 5$
35(b)	1.33×10^6	2	M1 for evaluating the absolute product $2^4 \times 3^3 \times 5^2 \times 7 \times 11 = 1330560$ A1 for correct rounding to 3 significant figures

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36(a)	15.75	2	M1 for fractional representations $\frac{7}{4}$ and $\frac{9}{4}$ or equivalent conversion to find $\text{LCM}(175, 225) = 1575$
36(b)	04:15	2	M1 for adding 15.75 minutes (15 minutes and 45 seconds) to 04:00 A1 for 04:15:45 or 04:16 rounded appropriately if required, or exact time 04:15 and 45 seconds
37(a)	50	2	M1 for 25×2 or for identifying $5^2 = 25$ or $\sqrt[3]{8} = 2$
37(b)	988	2	M1 for $1000 - 12$ or for identifying $10^3 = 1000$ or $\sqrt{144} = 12$
38(a)	38	2	M1 for $11 + 27 \times 1$ or for identifying $\sqrt{121} = 11$ or $3^3 = 27$
38(b)	26	2	M1 for $\dfrac{8 \times 13}{100 - 96}$ or $\dfrac{104}{4}$
39(a)	49	2	M1 for $4 \times 16 - 15$ or for identifying $\sqrt[3]{64} = 4$, $4^2 = 16$, or $\sqrt{225} = 15$
39(b)	151	2	M1 for $\dfrac{14}{2} + 144$ or for identifying $\sqrt{196} = 14$ or $12^2 = 144$
40(a)	-9	2	M1 for $2 \times 9 - 9 \times 3$ or $18 - 27$
40(b)	-105	2	M1 for $10 \times 2 - 125$ or $20 - 125$
41(a)	$\dfrac{7}{20}$	1	B1 for correct fraction simplified completely
41(b)	35%	1	B1
42(a)	$\dfrac{1}{8}$	1	B1 for correct fraction simplified completely
42(b)	2.6	1	B1
43(a)	$3\dfrac{1}{2}$	1	B1 for mixed number written in simplest form
43(b)	3.5	1	B1 ft from their part (a)
44(a)	8%	1	B1
44(b)	$\dfrac{2}{25}$	1	B1 for correct fraction simplified completely

Part	Answer	Marks	Partial Marks / Guidance
45(a)	$\frac{7}{5}$	1	B1 for improper fraction simplified completely
45(b)	1.4	1	B1
46(a)	$\frac{5}{8}$	1	B1 for correct fraction simplified completely
46(b)	4.6875	2	M1 for converting $\frac{11}{16}$ to 0.6875 via division or knowing $\frac{1}{16} = 0.0625$
47(a)	$\frac{13}{25000}$	2	M1 for $\frac{0.052}{100} = \frac{52}{100000}$ or equivalent fraction conversion steps
47(b)	$\frac{251}{40}$	2	M1 for converting elements to matching fractions or decimals: $5.45 - 0.375 + 1.20 = 6.275 \implies \frac{6275}{1000}$
48(a)	$\frac{27}{4}$	2	M1 for scaling decimals: $\frac{36 \times 1.5}{8} = \frac{54}{8}$ or converting to $\frac{9}{25} \times \frac{3}{2} \times \frac{25}{2}$
48(b)	675%	1	B1 ft from their part (a)
49(a)	$\frac{5}{8}$, 62.8%, $0.64 = \frac{16}{25}$	3	M1 for converting values to common decimal or percentage forms: $\frac{5}{8} = 0.625$, $62.8\% = 0.628$, $\frac{16}{25} = 0.64$ M1 for at least two values correctly ranked relative to each other
49(b)	$1\frac{9}{16}$	1	B1 for inverting to $\frac{25}{16}$ and expressing as a mixed number in its simplest form
50(a)	$\frac{1}{8} \times 12 \times \frac{1}{2} = \frac{12}{16} = \frac{3}{4} = 0.75 = 7.5 \times 10^{-1}$	2	M1 for converting 0.125 to $\frac{1}{8}$ or calculating numerator to 1.5 A1 for completely valid algebraic or numerical analytical trace showing the exact decimal reduction
50(b)	Any decimal d such that $0.875 < d < 0.88$ (e.g., 0.876)	2	M1 for recognizing $\frac{7}{8} = 0.875$ and $88\% = 0.88$

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51(a)	$\frac{5}{1}$ or 5	2	M1 for scaling expression by clearing decimals: $\frac{7.5\times3.2}{4.8}=\frac{24}{4.8}$ or alternative clean fractional tracking
51(b)	500%	1	B1 ft from their part (a)
51(c)	$\frac{1}{M}=\frac{1}{5}=0.2=2\times10^{-1}$, so $a=2$.	2	M1 for finding the reciprocal fraction $\frac{1}{5}$ and attempting decimal conversion A1 for explicit analytical verification showing $a=2$ with zero dots
52(a)	$\frac{13}{16}=\frac{13\times625}{16\times625}=\frac{8125}{10000}=0.8125$ or long division shown cleanly.	2	M1 for multiplying by a factor of $\frac{625}{625}$ or for demonstrating unambiguous long division A1 for complete proof of structural identity
52(b)	$A=B$, C, D or $\frac{13}{16}=0.8125$, 81.3%, $\frac{41}{50}$	3	M1 for converting values to common numeric platform: $C=0.8130$, $D=0.8200$ M1 for at least two values correctly ranked relative to each other
53(a)	$\frac{1}{2}$	3	M1 for evaluating numerator: $2.75-0.15=2.60$ M1 for evaluating denominator: $1.3\times4=5.2$ leading to $\frac{2.6}{5.2}$
53(b)	50%	1	B1 ft from their part (a)
53(c)	Any decimal d such that $0.5<d<0.501$ (e.g., 0.5005)	2	M1 for recognizing $P=0.5$ and $50.1\%=0.501$
54(a)	$\frac{2}{5}$, 0.42, $\frac{9}{20}$, 45%	2	M1 for converting numbers to a common form: $0.40, 0.42, 0.45, 0.45$ or equivalent percentages
54(b)	$>$	1	B1 for recognizing $-0.75>-0.80$
55(a)	$\frac{3}{8}$, 0.35, $\frac{7}{20}$, 32%	2	M1 for converting numbers to a common form: $0.375, 0.35, 0.32, 0.35$
55(b)	$\geq$ or $\leq$	1	B1 for recognizing both sides evaluate to 4, thus satisfying either inclusive boundary relation

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56(a)	$\frac{6}{5}$, 122%, $1.25 = \frac{5}{4}$	2	M1 for establishing decimal benchmarks: $1.25, 1.22, 1.25, 1.20$
56(b)	False	1	B1 for recognizing $\frac{1}{8}$ is exactly equal to 0.125
57(a)	0.08, $\frac{3}{40}$, $\frac{7}{100}$, 6.5%	2	M1 for evaluating numerical indices: $0.08, 0.07, 0.065, 0.075$
57(b)	$\frac{1}{3}$	1	B1 for converting values to decimals: $0.5, 0.4, 0.333\ldots$ to find the lowest
58(a)	$\frac{3}{5}$	2	M1 for establishing baseline forms: $0.545, 0.55, 0.60, 0.505$
58(b)	=	1	B1 for evaluating $\frac{14}{25} = \frac{56}{100} = 56\%$
59(a)	$\frac{5}{8}$, 62.8%, $\frac{13}{20}$, 0.66	2	M1 for converting to consistent forms: $0.625, 0.628, 0.65, 0.66$
59(b)	$=$	1	B1 for evaluating $4^{-2} = \frac{1}{16} = 0.0625$
60(a)	$\left(\frac{4}{9}\right)^{\frac{1}{2}}$, 66%, 0.65, $\frac{5}{8}$	2	M1 for evaluating index term to $\frac{2}{3}$ and finding matching decimal benchmarks: $0.666\ldots, 0.65, 0.66, 0.625$
60(b)	$=$	1	B1 for evaluating $-\frac{7}{5} = -1.4$
61(a)	67.5%, $\frac{17}{25}$, 0.682, $\left(\frac{1}{8}\right)^{-1}$ 2×10^{-1}	2	M1 for evaluating individual terms to equivalent forms: $0.68, 0.675, 0.682, 0.8$
61(b)	$=$ or $\geq$ or $\leq$	1	B1 for verifying both sides are structurally equivalent to 0.035
62(a)	$-\frac{7}{8}$	2	M1 for finding decimal absolute baselines: $0.75, 0.80, 0.82, 0.875$ to locate the largest distance from zero
62(b)	$>$	1	B1 for identifying $\left(\frac{27}{8}\right)^{-\frac{1}{3}} = \frac{2}{3} \approx 0.666\cdots > 0.6$
63(a)	$-\frac{7}{8}$	2	M1 for finding decimal absolute baselines: $0.75, 0.80, 0.82, 0.875$ to locate the largest distance from zero

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63(b)	$>$	1	B1 for identifying $\left(\frac{27}{8}\right)^{-\frac{1}{3}} = \frac{2}{3} \approx 0.666\cdots > 0.6$
64(a)	0.8485	1	B1
64(b)	C, B, A, D	2	B2 for all correct or B1 for 3 in correct relative order
64(c)	=	1	B1
65(a)	0.84	1	B1
65(b)	0.842	1	B1
65(c)	R, Q, P, S	2	B2 for all correct or B1 for 3 in correct relative order
66(a)	0.873	1	B1
66(b)	$\mathrm{V}_2, V_3, V_4, V_5, V_1$	2	B2 for all correct or B1 for 3 or 4 in correct relative order
66(c)	$>$	1	B1
67(a)	0.6283	1	B1
67(b)	P, K, N, M, L	2	B2 for all correct or B1 for 3 or 4 in correct relative order
67(c)	$>$	1	B1
68(a)	1.433	1	B1
68(b)	S, Q, R, P	2	B2 for all correct or B1 for 3 in correct relative order
68(c)	$>$	1	B1
69(a)	8	1	B1
69(b)	Q, S, P, R	2	B2 for all correct or B1 for 3 in correct relative order
69(c)	$<$	1	B1

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70(a)	$\left(\frac{27}{8}\right)^{\frac{2}{3}} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$	2	M1 for reciprocating fraction or taking cube root
70(b)	$\mathrm{X}_1, X_2, X_3, X_5, X_4$	2	B2 for all correct or B1 for 3 or 4 in correct relative order
70(c)	$>$	1	B1
71(a)	-5	1	B1 for -5
71(b)	11	1	B1 for 11
72(a)	$\frac{7}{10}$	1	B1 for $\frac{7}{10}$ or equivalent fraction e.g. $\frac{6}{10} + \frac{1}{10}$
72(b)	$\frac{8}{15}$	1	B1 for $\frac{8}{15}$ or M1 for $\frac{4 \times 2}{1 \times 15}$ after cancellation
73(a)	$\frac{1}{2}$	2	M1 for conversion to improper fractions: $\frac{7}{3} - \frac{11}{6}$ or $\frac{14}{6} - \frac{11}{6}$
73(b)	$3\frac{1}{3}$	2	M1 for $\frac{5}{4} \times \frac{8}{3}$ or $\frac{40}{12}$ or equivalent mixed number fraction
74(a)	5.25	1	B1 for 5.25
74(b)	0.024	1	B1 for 0.024
75(a)	23	2	M1 for correct order of operations leading to $5 - 3 \times (-8) - 6$ or $5 + 24 - 6$
75(b)	$-\frac{11}{7}$	2	M1 for numerator processing to -11 or denominator processing to 7
76(a)	$1\frac{23}{28}$	2	M1 for evaluating the multiplication first: $\frac{5}{3} \times \frac{6}{7} = \frac{10}{7}$
76(b)	$2\frac{2}{5}$	2	M1 for brackets evaluating to $\frac{32}{10}$ or $\frac{16}{5}$ and M1 for multiplying by $\frac{3}{4}$
77(a)	-11	2	M1 for $3 - 8 = -5$ or $5 \times (-5)$
77(b)	8.5	2	M1 for $1.2 \div 0.3 = 4$
78(a)	-24	1	A1 for correct value

Part	Answer	Marks	Partial Marks / Guidance
78(b)	$2\frac{3}{8}$ or $\frac{19}{8}$	2	M1 for $\frac{7}{4}+\frac{5}{8}=\frac{14}{8}+\frac{5}{8}$
79(a)	6	2	M1 for $3.5-1.1=2.4$ or $\frac{2.4}{0.4}$
79(b)	$\frac{1}{9}$	2	M1 for cancellation or $\frac{20}{180}$
80(a)	$(3+5)\times 2-4=12$	1	A1 for correct bracket placement
80(b)	$\frac{3}{10}$	2	M1 for positive sign or $\frac{6}{20}$
81(a)	-3	2	M1 for $3\times 4=12$ leading to $7-12+2$
81(b)	20	1	A1 for correct value
82(a)	3^6	1	B1
82(b)	5^3	1	B1
82(c)	2^{12}	1	B1
83(a)	1	1	B1
83(b)	$\frac{1}{16}$	1	B1
83(c)	$\frac{3}{8}$	1	B1
84(a)	5	1	B1
84(b)	3	1	B1
84(c)	8	2	B2 for correct answer, or M1 for $(\sqrt[4]{16})^3$ or 2^3 or $\sqrt[4]{4096}$
85(a)	2^2 or 4	1	B1
85(b)	6^3 or 216	1	B1
85(c)	$2\frac{1}{4}$	2	M1 for $\left(\frac{3}{2}\right)^2$ or $\frac{1}{\left(\frac{2}{3}\right)^2}$ or $\frac{2^{-2}}{3^{-2}}$ or $\frac{9}{4}$
86(a)	$\frac{1}{2}$	2	B2 for correct answer, or M1 for $\frac{1}{8^{\frac{1}{3}}}$ or $\frac{1}{2}$ seen in working

Part	Answer	Marks	Partial Marks / Guidance
86(b)	$\frac{2}{3}$	2	B2 for correct answer, or M1 for $\left(\frac{4}{9}\right)^{\frac{1}{2}}$ or $\frac{1}{\sqrt{\frac{9}{4}}}$ or $\frac{3}{2}$ seen
87(a)	4	2	M1 for $8^{\frac{2}{3}}$ or $(\sqrt[3]{8})^2$ or 2^2 or $\left(\frac{1}{4}\right)^{-1}$
87(b)	$\frac{5}{8}$	2	M1 for $125^{\frac{1}{3}} = 5$ or $2^{-3} = \frac{1}{8}$ or $\frac{5}{2^3}$
88(a)	-4	2	M1 for $3^x = 3^{-4}$ or $\frac{1}{81} = \frac{1}{3^4}$
88(b)	$\frac{7}{4}$	2	M1 for $(2^2)^{2x-1} = 2^5$ or $2(2x-1) = 5$ or $4x-2 = 5$
89(a)	12	3	M2 for $\frac{2^5 \times 2^3 \times 3^3}{3^2 \times 2^6}$ or M1 for breaking down bases into prime factors: $6^3 = 2^3 \times 3^3$ or $4^3 = 2^6$
89(b)	1	2	M1 for $\left(5^{\frac{1}{2}}\right)^4 = 5^2$ and $10^2 = 2^2 \times 5^2$ leading to $\frac{5^2 \times 2^2 \times 5^2}{2^2 \times 5^3}$ or $\frac{25 \times 100}{4 \times 125}$
90(a)	$\frac{27}{8}$	2	M1 for $\left(\frac{81}{16}\right)^{\frac{3}{4}}$ or $\left(\frac{3}{2}\right)^3$ or $\frac{2^{-3}}{3^{-3}}$
90(b)	$\frac{1}{17}$	3	M1 for $4^{\frac{3}{2}} = 8$, M1 for $27^{\frac{2}{3}} = 9$, then leading to $(8+9)^{-1}$
91(a)	$\frac{3}{5}$	2	M1 for converting to improper fraction $\left(\frac{25}{9}\right)^{-\frac{1}{2}}$ or finding $\sqrt{\frac{9}{25}}$
91(b)	4	2	M1 for writing all terms with base 3: $\frac{3^{-5} \times (3^2)^2}{(3^3)^{-1}}$ or $3^{-5+4-(-3)}$ or 3^2
92(a)	$x = 1, y = -2$	4	M1 for $x + 2y = -3$, M1 for $3x + y = 1$, M1 for attempting to solve the simultaneous linear system, A1 for both values correct.
93(a)	3×4^{-x}	2	M1 for $243^{\frac{x}{5}} = (3^5)^{\frac{x}{5}} = 3^x$ leading to $\frac{3^x \times 4^{-x}}{3^{x-1}}$ or $3^{x-(x-1)} \times 4^{-x}$
93(b)	0.0712	2	M1 for $3 \times 4^{-2.7}$ or $3 \times 0.0237...$ or $0.07119...$ seen

Part	Answer	Marks	Partial Marks / Guidance
94(a)	30 800	1	B1
94(b)	0.006 1	1	B1
95(a)	8×10^8	2	M1 for $2 \times 4 \times 10^{3+5}$ or 800,000,000
95(b)	3×10^{-4}	2	M1 for $3 \times 10^{2-6}$ or 0.0003
96(a)	1.5×10^3	2	M1 for 15×10^2 or 1500
97(a)	3×10^5	2	M1 for 30×10^4 or 300 000
97(b)	2.5×10^{-3}	2	M1 for 0.25×10^{-2} or $\frac{1}{400}$ or 0.0025
98(a)	3.6×10^5	2	M1 for $43 \times 10^4 - 7 \times 10^4$ or $430\,000 - 70\,000$ or 360 000
99(a)	-3	2	M1 for 0.009 or $(4.5 \times 10^{-5}) \times (2 \times 10^2) = 9 \times 10^{-3}$
99(b)	-6	2	M1 for $\frac{4}{1\,000\,000}$ or 0.000 004
100(a)	2.98×10^{-3}	2	M1 for $24 \times 10^{-4} + 5.8 \times 10^{-4}$ or $0.0024 + 0.00058$
100(b)	1.44×10^8	2	M1 for 1.44×10^k or 144 000 000
101(a)	7×10^2	3	M1 for numerator simplified to 14×10^4 or 1.4×10^5, M1 for division step $\frac{1.4 \times 10^5}{2 \times 10^2}$ or $\frac{140\,000}{200}$
102(a)	4.5×10^n	2	M1 for factorising out 10^{n-1} to get $(4 \times 10 + 5) \times 10^{n-1} = 45 \times 10^{n-1}$ or for equating powers $4 \times 10^n + 0.5 \times 10^n$
102(b)	$3.2 \times 10^{n+1}$	2	M1 for $\frac{16 \times 10^{2n}}{5 \times 10^{n-1}}$ or for $3.2 \times 10^{2n-(n-1)}$
103(a)	5×10^4	3	M1 for converting numerator to same power: $12 \times 10^{-4} + 8 \times 10^{-4} = 20 \times 10^{-4}$ or 2×10^{-3}, M1 for dividing by 4×10^{-8} to get 0.5×10^5
104(a)	-1	3	M1 for $(6.25 \times 10^{2k}) \times (8 \times 10^3)$, M1 for $50 \times 10^{2k+3} = 5 \times 10^{2k+4}$, A1 for equating $2k+4 = 2$ to find $k = -1$

Part	Answer	Marks	Partial Marks / Guidance
104(b)	8×10^{-15}	2	M1 for $\frac{1}{125 \times 10^{12}}$ or 0.008×10^{-12}
105(a)	2.1×10^{-5}	2	M1 for $30 \times 10^{-6} - 9 \times 10^{-6} = 21 \times 10^{-6}$ or $3 \times 10^{-5} - 0.9 \times 10^{-5}$
105(b)	1×10^{-2}	3	M1 for $X^2 = 9 \times 10^{-10}$, M1 for $\frac{9 \times 10^{-10}}{9 \times 10^{-6}} = 1 \times 10^{-4}$ or 10^{-6} error in division, A1 for $\sqrt[3]{10^{-6}} = 10^{-2}$
106(a)	2.5×10^{-1}	2	M1 for numerator $= 25 \times 10^{-6}$ and denominator $= 1 \times 10^{-4}$ leading to 25×10^{-2}
106(b)	2.44×10^{-4}	3	M1 for $\frac{1}{M} = 4100$ or 4.1×10^3, M1 for $M = \frac{1}{4100} = \frac{1}{4.1} \times 10^{-3}$ or showing step leading to 0.0002439... or $\frac{10}{41} \times 10^{-4}$
107(a)	4.39×10^7	2	M1 for $42 \times 10^6 + 1.9 \times 10^6$ or $4.2 \times 10^7 + 0.19 \times 10^7$ or 43 900 000
107(b)	8.78×10^3	2	M1 for $\frac{4.39 \times 10^7}{5 \times 10^3}$ or 8780
108(a)	5.73×10^8	2	M1 for $5.82 \times 10^8 - 0.094 \times 10^8$ or $582 \times 10^6 - 9.4 \times 10^6$ or 572 600 000
108(b)	1.62%	2	M1 for $\frac{9.4 \times 10^6}{5.82 \times 10^8} \times 100$ or $0.016151... \times 100$
109(a)	30	2	M1 for $\frac{7.5 \times 10^{11}}{2.5 \times 10^{10}}$ or $\frac{75 \times 10^{10}}{2.5 \times 10^{10}}$
109(b)	3.88×10^{11}	2	M1 for $\frac{7.5 \times 10^{11} + 0.25 \times 10^{11}}{2}$ or $\frac{7.75 \times 10^{11}}{2}$ or 387 500 000 000
110(a)	4.85×10^{14}	2	M1 for $4.76 \times 10^{14} + 0.0895 \times 10^{14}$ or 484.95×10^{12} or 484 950 000 000 000
110(b)	53.2	2	M1 for $\frac{4.76 \times 10^{14}}{8.95 \times 10^{12}}$ or 53.1843...
111(a)	6.23×10^2	3	M1 for multiplying numerator to 7.7872×10^4, M1 for dividing by 1.25×10^2 to get 622.976, A1 for standard form to 3 sig figs
111(b)	623	1	B1 ft from part (a)
112(a)	483.71	1	B1

Part	Answer	Marks	Partial Marks / Guidance
112(b)	484	1	B1
113(a)	1000	3	M1 for writing individual values to 1 sig fig as 40, 5, and 0.2, M1 for $\frac{40 \times 5}{0.2}$ or $\frac{200}{0.2}$, A1 for 1000.
114(a)	0.0030	1	B1
114(b)	80 000	1	B1
115(a)	10	2	M1 for $\frac{40 \times 0.05}{\sqrt{0.04}}$ or $\frac{2}{0.2}$
115(b)	0.05	1	B1
116(a)	0.06	2	M1 for showing individual 1 sig fig roundings: $\frac{6 \times 0.2}{200}$ or $\frac{1.2}{200}$ or $\frac{12}{2000}$
116(b)	$\frac{3}{50}$	1	B1
117(a)	£70	2	M1 for $\frac{140}{2+3+5} \times 5$ oe
117(b)	£42	2	M1 for $(5-2) \times 14$ or $70-28$ oe
118(a)	300 g	2	M1 for $\frac{120}{8} \times 20$ or 15×20
118(b)	30	2	M1 for $\frac{450}{15}$ oe
119(a)	8 kg	2	M1 for $\frac{14}{7} \times 4$ oe
119(b)	42 kg	2	M1 for $3.5 \times (7+4+1)$ oe
120(a)	600 ml	2	M1 for $\frac{3}{5+3+1} \times 1800 = \frac{3}{9} \times 1800$
120(b)	5 : 3 : 2	2	M1 for doubling the last component to get 5 : 3 : 2 visible or structured
121(a)	$\frac{3}{10}$	1	Correct answer only
121(b)	60 kg	3	M1 for $7-3=4$ parts, M1 for 1 part $= \frac{24}{4} =$ 6 kg, A1 for $10 \times 6 = 60$ kg
122(a)	20 cm	2	M1 for $\frac{3000}{150}$ oe
122(b)	900 m^2	3	M1 for 400×150^2, M1 for division by 10000 to convert cm^2 to m^2

Part	Answer	Marks	Partial Marks / Guidance
123(a)	3 : 8	2	M1 for 45 : 120 or correct conversion of units to common form
124(a)	$\frac{7}{9}$	1	Correct answer only
124(b)	Proof	2	M1 for expressing salt as $\frac{2}{9}V$ and new water as $\frac{7}{9}V+60$, M1 for equating new water to $4 \times$ salt $= 4\left(\frac{2}{9}V\right) = \frac{8}{9}V$
124(c)	540 ml	2	M1 for isolating $\frac{1}{9}V = 60$ or equivalent shift
125(a)	$n = 200$	3	M1 for actual area $= 12 \times 16 = 192$ m^2, M1 for converting to cm^2: $192 \times 10000 = 1920000$ cm^2, A1 for $n^2 = \frac{1920000}{48} = 40000 \implies n = 200$
125(b)	27 cm	2	M1 for $\frac{5400}{200}$ oe
126(a)	£1200	2	M1 for $\frac{4}{12} \times 3600$ oe
126(b)	Manager C, loss of £100	3	M1 for finding new shares: $A = 800, B = 1200, C = 1600$, M1 for comparing old shares ($A = 900, B = 1200, C = 1500$) to identify A loses £100 and C gains £100, A1 for Manager A with £100 loss
127(a)	3 hours 30 minutes	2	M1 for $\frac{1400}{400}$ or 3.5 hours
127(b)	600	2	M1 for $\frac{1400}{2\frac{1}{3}}$ or $\frac{1400 \times 3}{7}$
128(a)	5.4	3	M1 for $15 \times 60 = 900$ seconds, M1 for $6 \times 900 = 5400$ m
128(b)	$\frac{6 \times 3600}{1000} = 21.6$	2	M1 for 6×3600 or $\frac{6}{1000}$ seen
129(a)	12	2	M1 for $\frac{45}{3\frac{3}{4}}$ or $\frac{45 \times 4}{15}$
129(b)	2.7	2	M1 for $45 \times \frac{6}{100}$ or $\frac{270}{100}$

Part	Answer	Marks	Partial Marks / Guidance
130(a)	$18\frac{4}{7}$	3	M1 for total distance $=36$, M1 for total time $=\frac{12}{16}+\frac{24}{20}=\frac{3}{4}+\frac{6}{5}=\frac{39}{20}$ hours, A1 for $\frac{36\times 20}{39}=\frac{240}{13}$ or $18\frac{6}{13}$
130(b)	4.6	2	M1 for mass $=4000\times 1.15=4600$ g
131(a)	265	2	M1 for distance of second part $=100\times 1\frac{3}{4}=$ 175 km
131(b)	106	2	M1 for total time $=\frac{90}{120}+1\frac{3}{4}=\frac{3}{4}+\frac{7}{4}=2.5$ hours, then $\frac{265}{2.5}$
132(a)	$\frac{2x}{\frac{x}{60}+\frac{x}{90}}=72$	3	M1 for setting total distance as $2x$, M1 for total time expression $\frac{x}{60}+\frac{x}{90}=\frac{5x}{180}=\frac{x}{36}$, A1 for $\frac{2x}{\frac{x}{36}}=72$
132(b)	120	2	M1 for total distance $=72\times 3\frac{1}{3}=72\times\frac{10}{3}=$ 240 km, then $\frac{240}{2}$
133(a)	1620	1	B1 for 1620
133(b)	751.68	2	M1 for $\frac{2}{5}\times 1620=648$, then 648×1.16
133(c)	29	2	M1 for fuel volume $=\frac{250}{100}\times 8=20$ litres, then 20×1.45
134(a)	$\frac{2v(v-15)}{(v-15)+v}=\frac{2v^2-30v}{2v-15}$	3	M1 for total distance $=2d$ and time expressions $\frac{d}{v}$ and $\frac{d}{v-15}$
134			M1 for adding times to get $\frac{d(v-15)+dv}{v(v-15)}=\frac{d(2v-15)}{v(v-15)}$
134(b)	60	3	M1 for $\frac{2v^2-30v}{2v-15}=40\Rightarrow 2v^2-30v=80v-600$
134			M1 for $2v^2-110v+600=0\Rightarrow(2v-20)(v-60)=0$

Part	Answer	Marks	Partial Marks / Guidance
135(a)	$\dfrac{d}{\dfrac{\left(\frac{d}{3}\right)}{x}+\dfrac{\left(\frac{2d}{3}\right)}{2x}}=\dfrac{3}{2}x$	3	M1 for total time $=\dfrac{d}{3x}+\dfrac{2d}{6x}$
135			M1 for simplifying denominator to $\dfrac{2d}{3x}$
135			A1 for $\dfrac{d}{\left(\frac{2d}{3x}\right)}=\dfrac{3x}{2}$
135(b)	48	2	M1 for equating $\dfrac{3}{2}x=\dfrac{270}{3\frac{3}{4}}=\dfrac{270\times4}{15}=72$
136(a)	1 hour 30 minutes	2	M1 for $\dfrac{114}{76}$
136(a)(i)	74.7 km/h	3	M1 for total time $=1.5+2.3=3.8$ hours
136(a)(ii)			M1 for $\dfrac{284}{3.8}$
137(a)	EUR 522.90	2	M1 for 450×1.162
137(a)(i)	62.0 m/s	3	M1 for total seconds $=82\times60=4920$ s
137(a)(ii)			M1 for $\dfrac{305000}{4920}$
138(a)	CAD 6753	3	M1 for $3850\times1.284=4943.4$ USD
138			M1 for $\dfrac{4943.4}{0.732}$
138			A1 for 6753.28 rounded to 6753
138(b)	GBP 89.10	4	M1 for fuel volume $=\dfrac{645}{100}\times14.8=$ 95.46 litres
138			M1 for cost in CAD $=95.46\times1.64=$ 156.5544 CAD
138			M1 for converting CAD to GBP: $\dfrac{156.5544\times0.732}{1.284}$
138			A1 for 89.10 (allow 89.09 to 89.11)
139(a)	280 GBP	2	M1 for $250\times\dfrac{112}{100}$ or $250+30$

Part	Answer	Marks	Partial Marks / Guidance
139(b)	252 GBP	2	M1 for their $280 \times \frac{90}{100}$ or their $280 - 28$
140(a)	150 EUR	3	M2 for $180 \div \frac{120}{100}$ or $180 \times \frac{100}{120}$ or M1 for identifying $120\% = 180$
141(a)	605 grams	3	M2 for $500 \times \left(\frac{110}{100}\right)^2$ or $500 \times 1.1 \times 1.1$ or M1 for $500 \times 1.1 = 550$
141(b)	121%	1	B1 for 121% or their $\frac{605}{500} \times 100$
142(a)	1000 GBP	2	M1 for $800 \times \left(1 + \frac{25}{100}\right)$ or $800 \times \frac{5}{4}$
142(a)(i)	6.25%	3	M1 for their $1000 \times \frac{85}{100}$ or 850
142(a)(ii)			M1 for $\frac{\text{their } 850 - 800}{800} \times 100$ or $\frac{5}{4} \times \frac{85}{100} = \frac{425}{400} = 1.0625$
143(a)	2646 EUR	2	M1 for Year 1: $2400 \times 1.05 = 2520$
143(a)(i)			M1 for Year 2: their $2520 \times 1.05 = 2646$ with no dots on line
143(a)(ii)	132.30 EUR	2	M1 for $2778.30 - 2646$ or $2646 \times \frac{5}{100}$
144(a)	$\left(1 + \frac{a}{100}\right)\left(1 - \frac{a}{100}\right) = \frac{24}{25} \Rightarrow 1 - \left(\frac{a}{100}\right)^2 = \frac{24}{25}$	3	M1 for product of multipliers or $P\left(1 - \frac{a^2}{10000}\right)$
144(a)(i)	$\Rightarrow \left(\frac{a}{100}\right)^2 = 1 - \frac{24}{25} = \frac{1}{25}$		M1 for clear algebraic simplification to given form
144(a)(ii)	20	1	B1 for 20 (reject -20)
144(a)(iii)	50000 GBP	2	M1 for $48000 \div \frac{24}{25}$ or $48000 \times \frac{25}{24}$
145(a)	1039.50 GBP	2	M1 for $4500 \times 0.0385 \times 6$
145(b)	5539.50 GBP	1	FT for $4500 +$ their (a)
146(a)	141000 USD	3	M2 for $125000 \times (1.0425)^3$ or M1 for 125000×1.0425
146(b)	16400 USD	1	FT for their (a) $- 125000$
147(a)	18742.66 EUR	2	M1 for 14500×1.0435^6

Part	Answer	Marks	Partial Marks / Guidance
147(b)	18030.44 EUR	2	M1 for Ans from (a) $\times (1-0.038)$
147(c)	124.3% or 124%	2	M1 for $\frac{\text{Ans from (b)}}{14500} \times 100$
148(a) **148**	196620 AUD	3	M1 for $t = 7.25$ M1 for $480000 \times 0.0565 \times 7.25$
148(b)	Show derived statement	1	A1 for explicit statement showing $1+\frac{4.95}{100} = 1.0495$ applied compoundingly
148(c) **148**	11 years	2	M1 for systematic trial or setting up equation $370000 \times 1.0495^n > 600000$ A1 for $n = 11$ (since 10 years gives ≈ 599650)
149(a)	06 15	2	M1 for bridging midnight, e.g., 22 45 + 1 hour 15 minutes is 00 00, leaving 6 hours 15 minutes.
149(b)	Wednesday	1	B1 for the correct day following midnight.
150(a)	18 15	1	B1 for correct addition of 4 hours.
150(b)	110 minutes	2	M1 for 1 hour 50 minutes or $60+50$.
151(a)	13 20	1	B1 for subtracting 2 hours from departure time.
151(b)	18 20 on 13 April	2	M1 for adding 3 days to the date (13 April) and 5 hours to the time ($1320+5$ hours).
152(a)	10:45 p.m.	1	B1 for correct time format with p.m.
152(b)(i)	19 20	2	M1 for adding 9 hours and 11 hours 35 minutes to 22 45 or equivalent
152(b)(ii)	Wednesday	1	B1 for Wednesday
153(a)	Tuesday 20 20	2	M1 for correct time zone shift to departure local time ($2145+9$ hours $=$ Tuesday 0645) or adding duration first.
153(b)	13.6	2	M1 for $\frac{35}{60}$ seen or 13.583...

Part	Answer	Marks	Partial Marks / Guidance
154(a)	$6\sqrt{2}$	2	M1 for 6^2+6^2
154(b)	$3\sqrt{2}$	1	
154(c)	Fully correct proof	2	M1 for $AE^2=(3\sqrt{2})^2+4^2$
155(a)	5	2	M1 for $\sqrt{3^2+4^2}$
155(b)	13	2	M1 for $\sqrt{5^2+12^2}$
155(c)	$\sin(\angle CAG)=\frac{CG}{AG}$	1	M1 for identifying the correct angle $\angle CAG$
155	$\sin(\angle CAG)=\frac{12}{13}$	1	A1 for concluding the exact fractional proof
156(a)	$6\sqrt{2}$	2	M1 for $\sqrt{6^2+6^2}$
156(b)	$3\sqrt{2}$	1	B1 for $\frac{1}{2}\times 6\sqrt{2}$
156(c)	45°	2	M1 for $\tan(\angle VPM)=\frac{3\sqrt{2}}{3\sqrt{2}}$
157(a)	$2\sqrt{2}$	2	M1 for $\sqrt{2^2+2^2}$
157(b)	$2\sqrt{3}$	2	M1 for $\sqrt{2^2+(2\sqrt{2})^2}$
157(c)	$\cos(\angle DFH)=\frac{FH}{DF}$	1	M1 for identifying the correct angle $\angle DFH$
157	$=\frac{2\sqrt{2}}{2\sqrt{3}}=\sqrt{\frac{2}{3}}$	1	A1 for concluding the exact fractional proof
158(a)	$5\sqrt{3}$	2	M1 for $\sqrt{5^2+(5\sqrt{2})^2}$
158(b)	10	2	M1 for $\sqrt{5^2+(5\sqrt{3})^2}$
158(c)	30°	2	M1 for $\sin(\angle CDB)=\frac{5}{10}$ or $\tan(\angle CDB)=\frac{5}{5\sqrt{3}}$
159(a)	10	2	M1 for $\sqrt{8^2+6^2}$
159(b)	5	1	B1 for $\frac{1}{2}\times 10$
159(c)	60°	2	M1 for $\tan(\angle VYM)=\frac{5\sqrt{3}}{5}$
160(a)	5	2	M1 for $AC=\sqrt{8^2+6^2}$ or 10
160(b)	13	2	M1 for $\sqrt{5^2+12^2}$

Part	Answer	Marks	Partial Marks / Guidance
160(c)	Correct analytical proof	2	M1 for identifying $\angle VAM$ as the required angle M1 for stating $\cos(\angle VAM) = \frac{5}{13}$
161(a)	5	2	M1 for $\sqrt{3^2+4^2}$
161(b)	13	2	M1 for $\sqrt{5^2+12^2}$ or $\sqrt{3^2+4^2+12^2}$
161(c)	$\sin(\angle GAC) = \frac{12}{13}$	2	M1 for identifying $\angle GAC$ as the correct angle
161			A1 for clear working showing $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{CG}{AG} = \frac{12}{13}$ without erroneous steps
162(a)	$2x^2$	1	B1 for x^2+x^2 or $2x^2$
162(b)(i)	proof	2	M1 for $AG^2 = AC^2 + CG^2$ or $3x^2 = 6^2$
162			A1 for $x^2 = 12$ leading to $x = \sqrt{12} = 2\sqrt{3}$
162(b)(ii)	$\frac{\sqrt{3}}{3}$	2	M1 for identifying $\sin(\angle GAC) = \frac{CG}{AG}$
162			A1 for $\frac{2\sqrt{3}}{6}$ equivalent to $\frac{\sqrt{3}}{3}$ or $\frac{1}{\sqrt{3}}$
163(a)	$6\sqrt{2}$	2	M1 for $AB^2 = 6^2+6^2$
163(b)	$OM^2 + (3\sqrt{2})^2 = 6^2$	M1	For correct Pythagorean setup in the base circle
163	$OM^2 = 36-18 = 18 \implies OM = 3\sqrt{2}$	A1	For correct isolation and simplification to $3\sqrt{2}$
163(c)	$\frac{4\sqrt{2}}{3}$	2	M1 for $\tan(\angle PMO) = \frac{8}{3\sqrt{2}}$
164(a)	10	2	M1 for $AC^2 = 8^2+6^2$
164(b)	$DC^2 = 24^2+10^2$	M1	For correct Pythagorean setup using vertical height and AC
164	$DC^2 = 576+100 = 676 \implies DC = 26$	A1	For correct evaluation ending at 26
164(c)	$DB^2 = 24^2+8^2 = 640$	M1	For calculating DB^2

Part	Answer	Marks	Partial Marks / Guidance
164	$DB^2 + BC^2 = 640 + 36 = 676$	M1	For summing squares of the legs in $\triangle DBC$
164	$676 = DC^2$, hence right-angled	A1	For conclusive statement showing $DB^2 + BC^2 = DC^2$
164(d)	$\frac{3}{13}$	1	
165(a)	Two correct points plotted as crosses at $(35, 205)$ and $(74, 380)$	2	B1 for one correct point plotted accurately
165(b)	Positive	1	
165(c)	$y = 4.57x + 46.6$	2	B1 for $a = 4.57$ or $b = 46.6$ (accept up to 3 sig figs)
165(d)(i)	595	2	M1 for substituting $x = 120$ into their regression equation
165(d)(ii)	Unreliable because it is outside the range of the given data (extrapolation)	1	Accept equivalent language indicating extrapolation is invalid
166(a)	Points correctly plotted at $(16, 210)$ and $(26, 390)$ as crosses	2	B1 for each correctly plotted point (± 1 small square).
166(b)	Positive	1	B1 for Positive. Accept strong positive.
166(c)	$y = 17.5x - 64.4$	2	B1 for $m = 17.5$. B1 for $c = -64.4$ (or -64.375).
166(d)	286	2	M1 for substitution $17.5 \times 20 - 64.375$. A1 for 285 to 286.
166(e)	Reliable, it is interpolation (or inside the range of given data).	1	B1 for identifying interpolation or within data range.
167(a)	Points correctly plotted at $(80, 38)$ and $(200, 25)$ as crosses	2	B1 for each correctly plotted point (± 1 small square).
167(b)	Negative	1	B1 for Negative. Accept strong negative.

Part	Answer	Marks	Partial Marks / Guidance
167(c)	$y = -0.110x + 47.2$	2	B1 for $m = -0.110$. B1 for $c = 47.2$ (or 47.19).
167(d)	14.2	2	M1 for substitution $-0.110 \times 300 + 47.19$. A1 for 14.1 to 14.3.
167(e)	Extrapolation, as 300 km is beyond the range of recorded distances.	1	B1 for identifying extrapolation and referencing the data bounds.
168(a)	Points correctly plotted at $(25, 28)$ and $(35, 36)$ as crosses	2	B1 for each correctly plotted point (± 1 small square).
168(b)	Positive	1	B1 for Positive. Accept strong positive.
168(c)	$y = 0.843x + 7.04$	2	B1 for $m = 0.843$ (or $0.8428\ldots$). B1 for $c = 7.04$ (or $7.035\ldots$).
168(d)	22.2	2	M1 for substitution $0.843 \times 18 + 7.04$. A1 for 22.1 to 22.3.
168(e)	It is interpolation because 18 g lies within the data set range.	1	B1 for reference to interpolation or being inside the known range.
169(a)	Two points correctly plotted at $(5, 27)$ and $(11, 52)$ as crosses	2	B1 for each correctly plotted point
169(b)	Positive correlation	1	
169(c)	$y = 4.32x + 5.79$	2	B1 for $m = 4.32$ (4.315...), B1 for $c = 5.79$ (5.793...)
169(d)	48.9 or 49	2	M1 for $4.315 \times 10 + 5.793$
169(e)	16 kg is outside the range of the given data / it is extrapolation	1	Accept equivalent reasonable explanations
170(a)	Plotting $(25, 17)$ and $(60, 8)$ as crosses	2	B1 for each correct point plotted exactly on lattice.
170(b)	Negative	1	Accept strong negative.

Part	Answer	Marks	Partial Marks / Guidance
170(c)	$y = -0.226x + 21.9$	2	B1 for $a = -0.226$ and B1 for $b = 21.9$.
170(d)	14.0 thousand GBP	1	M1 for $-0.226 \times 35 + 21.9$.
170(e)	120 is outside data range / value becomes negative	1	B1 for identifying extrapolation or impossible negative GBP value.
171(a)	Plotting $(9,58)$ and $(15,82)$ as crosses	2	B1 for each correctly plotted point.
171(b)	Positive	1	Accept strong positive.
171(c)	$y = 3.35x + 27.9$	2	B1 for $a = 3.35$ and B1 for $b = 27.9$.
171(d)	61.4	1	M1 for $3.35 \times 10 + 27.9$ evaluated. Accept 61.
171(e)	Invalid, extrapolation	1	B1 for stating it is extrapolation or outside the data range.
172(a)	Diagram C	1	
172(b)	Diagram B	1	
172(c)	Strong positive correlation	1	Accept 'positive correlation'
173(a)(i)	12 000 000	1	B1
173(a)(ii)	12 000	1	B1
173(b)	4500	1	B1
174(a)(i)	60 000	1	B1
174(a)(ii)	0.06 or $\frac{3}{50}$	1	B1
174(b)	0.35 or $\frac{7}{20}$	1	B1
175(a)(i)	50 000	1	B1
175(a)(ii)	1.5 or $\frac{3}{2}$	1	B1
175(b)	0.08 or $\frac{2}{25}$	1	B1
176(a)(i)	2800	1	B1

Part	Answer	Marks	Partial Marks / Guidance
176(a)(ii)	0.25 or $\frac{1}{4}$	1	B1
176(b)	450	1	B1 for converting to 450 ml or recognising $1\text{ l} = 1000\text{ cm}^3$
177(a)	$350\,000\text{ mm}^3$	1	B1 for 350×1000
177(b)	6.755 kg	2	M1 for $350 \times 19.3 = 6755$ g and dividing by 1000
177(c)	2.8 cm^3	2	M1 for $350 \times \left(\frac{1}{5}\right)^3$ or $\frac{350}{125}$
178(a)	9 litres per minute	2	M1 for $\frac{150 \times 60}{1000}$ or equivalent
178(b)	Show that height increases by 3.6 mm	3	M1 for converting volume per hour: $9 \times 60 = 540$ litres or 0.54 m^3
178			M1 for height $= \frac{\text{volume}}{\text{area}} = \frac{0.54}{2.5} = 0.216$ m
178			A1 for converting 0.216 m to 216 mm and dividing by 60 to get 3.6 mm or equivalent complete proof
179(a)	27 litres	1	B1 for 0.027×1000
179(b)	30 cm	2	M1 for $\sqrt[3]{0.027} = 0.3$ m or $\sqrt[3]{27\,000\text{ cm}^3}$
179(c)	10 minutes	2	M1 for $\frac{27\,000\text{ ml}}{45\text{ ml/s}} = 600$ seconds and dividing by 60
180(a)	4×10^4 mm	2	M1 for converting to 40 m or multiplying by 10^6
180(b)	1600 m^2	2	M1 for $(40\text{ m})^2$ or converting 0.04^2 km^2 into m^2
180(c)	240 litres	2	M1 for $1600 \times 150 = 240000$ ml or multiplying by $\frac{150}{1000}$
181(a)	Negative correlation	1	B1 for identifying negative gradient / correlation.
181(b)	75.4	1	B1 for $92.5 - 0.38(45) = 75.4$.

Part	Answer	Marks	Partial Marks / Guidance
181(c)	75.0 metres	2	M1 for $64.0 = 92.5 - 0.38x$. A1 for $x = 75$ or 75.0.
182(a)	Correctly plots $(4.0, 102500)$ and $(7.0, 62500)$.	2	B1 for each correct point plotted accurately on/between gridlines.
182(b)	Negative	1	B1 for Negative.
182(c)	$y = -13100x + 155000$	2	B2 for $-13143x + 155476$ or proper 3 sf equivalent.
182			M1 for finding slope $m = -13142.85...$ and $c = 155476.19...$
182(d)	70000	2	M1 for calculation $y = -13143(6.5) + 155476$. Accept reasonable answers based on rounded equation.
183(a)	Negative	1	B1 for Negative.
183(b)(i)	137	2	M1 for substituting $\bar{x} = 16$ into the equation: $y = 342.7 - 12.85(16)$.
183(b)(ii)	85.7	2	M1 for substitution: $y = 342.7 - 12.85(20)$.
183(c)	15	3	M1 for setting equation: $150 = 342.7 - 12.85x$.
183			M1 for solving for x: $x = \frac{192.7}{12.85} = 14.996...$
183			A1 for rounding correctly to nearest whole number.
183(d)	No, because 35 is outside the range of collected data (extrapolation).	1	B1 for stating 'No' with a valid reason involving extrapolation or range limits.
184(a)	$50000 \times 0.8^2 = 32000$	2	M1 for 50000×0.8^2 or $50000 \times \left(\frac{4}{5}\right)^2$ or 40000 seen
184(b)	25600 EUR	2	M1 for 32000×0.8 or 50000×0.8^3
185(a)	1260	1	B1 for 1260

Part	Answer	Marks	Partial Marks / Guidance
185(b)	1323	2	M1 for 1260×1.05 or 1200×1.05^2
186(a)	2000 AUD	2	M1 for $16000 \times \left(\frac{1}{2}\right)^3$ or 8000 then 4000 seen
186(b)	14000 AUD	1	B1 for 16000 – ans from (a)
187(a)	654	2	M1 for 480×1.064^5 or 654.49...
187(b)	12	3	M1 for setting up inequality or equation $480 \times 1.064^t > 1000$ or $1.064^t > \frac{1000}{480}$
187			A1 for 11.8... or trial and error showing $t = 11 \to 952$ and $t = 12 \to 1013$
188(a)	43945	2	M1 for $75000 \times (1 - 0.125)^4$ or 75000×0.875^4
188(b)	41.4%	2	M1 for $\frac{75000 - 43945.31}{75000} \times 100$ or $1 - 0.875^4$
189(a)	58900	2	M1 for 42000×1.058^6 or 58862.6...
189(b)	10	3	M1 for $1.058^t \geq 1.75$ or $42000 \times 1.058^t \geq 73500$
189			A1 for 9.92... or trial and error showing $t = 9 \to 69707$ and $t = 10 \to 73750$
190(a)	47600	2	M1 for 12500×1.182^8 or 47602.8...
190(b)	10	3	M1 for $1.182^t = 5$ or $12500 \times 1.182^t = 62500$
190			A1 for 9.62... or trial and error showing $t = 9 \to 56265$ and $t = 10 \to 66505$
191(a)	1993	2	M1 for 1420×1.043^8 or 1992.59...
191(b)	17	3	M1 for $1420 \times 1.043^t = 2840$ or $1.043^t = 2$
191			A1 for $t = 16.46...$
191			A1 for 17
192(a)	216000 EUR	2	M1 for $485000 \times (1 - 0.126)^6$ or 216086.41...

Part	Answer	Marks	Partial Marks / Guidance
192(b)	Value after 5 years $= 485000 \times 0.874^5 =$ 247238.46... EUR		
192	Value after 6 years $= 216086.41...$ EUR		
192	Half of initial value $=$ 242500 EUR		
192	Since $216086.41 < 242500 < 247238.46$, it falls below half during the 6th year.	3	M1 for finding value at $t = 5$
192			M1 for comparing both values to $\frac{485000}{2} =$ 242500
192			A1 for complete correct logical explanation
193(a)	2400	3	M1 for 1.5 years $=$ 18 months
193			M1 for $8500 \times (1 - 0.068)^{18}$ or 2415.54...
193			A1 for 2400 (nearest hundred)
193(b)	18	3	M1 for $8500 \times 0.932^m < 2500$ or $0.932^m < \frac{5}{17}$
193			A1 for $m = 17.36...$
193			A1 for 18
194(a)	41100	2	M1 for 45000×0.982^5
194(b)	Fully correct working	2	M1 for $32000 \times 1.024^n > 45000 \times 0.982^n$
194(c)	9	3	M2 for $n > \frac{\log(1.40625)}{\log(1.0427...)}$ or M1 for evaluated trial and error yielding 8 and 9
195(a)	76765.63 or 76800	2	M1 for 125000×0.85^3
195(b)	11.5	3	M2 for $1 - \frac{r}{100} = \left(\frac{41500}{76765.625}\right)^{\frac{1}{5}}$ or M1 for $76765.625 \times \left(1 - \frac{r}{100}\right)^5 = 41500$

Part	Answer	Marks	Partial Marks / Guidance
195(c)	60600	3	M2 for $\frac{28160}{0.88^6}$ or M1 for $V \times 0.88^6 = 28160$
196(a)	1249	2	M1 for 850×1.08^5
196(b)	10.5	3	M2 for $k = 100 \times \left(1 - \left(\frac{720}{1249}\right)^{\frac{1}{5}}\right)$ or M1 for $1249 \times \left(1 - \frac{k}{100}\right)^5 = 720$
196(c)	14	4	M2 for $t > \frac{\log(500 \div 1249)}{\log(0.895)}$ and M1 for adding 5 to the derived integer value of t
197(a)	5.39	3	M2 for $\left(\left(\frac{18500}{15000}\right)^{\frac{1}{4}} - 1\right) \times 100$ or M1 for $15000 \times \left(1 + \frac{r}{100}\right)^4 = 18500$
197(b)	Fully correct working	3	M1 for 15000×1.0539^7 evaluated as 21667 and M1 for $V \times 0.965^7 = 10833.5$
197(c)	12	4	M2 for $t > \frac{\log(2.78)}{\log(1.092)}$ or M1 for setting up inequality $15000 \times 1.0539^t > 3 \times 13899 \times 0.965^t$
198(a)	$4\sqrt{2}$	1	
198(b)	$2\sqrt{6}$	2	M1 for $\frac{12\sqrt{6}}{6}$
199(a)	$7\sqrt{3}$	2	B1 for $5\sqrt{3}$ or $2\sqrt{3}$
199(b)	$2\sqrt{5}$	2	M1 for $\frac{10\sqrt{5}}{5}$
200(a)	$4\sqrt{5}$	1	
200(b)	$\frac{15\sqrt{3}}{3}$	M1	
200	$5\sqrt{3}$	A1	
201(a)	$\sqrt{7}$	2	M1 for $2\sqrt{7}$
201(b)	$4\sqrt{2}$	2	M1 for $\frac{8\sqrt{2}}{2}$
202(a)	$10\sqrt{3}$	2	M1 for $12\sqrt{3}$ or $2\sqrt{3}$

Part	Answer	Marks	Partial Marks / Guidance
202(b)	$7-3\sqrt{3}$	3	M1 for multiplying numerator and denominator by $2-\sqrt{3}$
202			M1 for numerator $10-5\sqrt{3}+2\sqrt{3}-3$
203(a)	$2\sqrt{2}$	2	M1 for $7\sqrt{2}-5\sqrt{2}$
203(b)	$6-2\sqrt{5}$	3	M1 for multiplying numerator and denominator by $3-\sqrt{5}$
203			M1 for denominator $9-5$ or 4
204(a)	$4+\sqrt{5}$	2	M1 for $10+4\sqrt{5}-3\sqrt{5}-6$
204(b)	$3\sqrt{2}$	2	M1 for $\frac{12\sqrt{8}}{8}$ or $\frac{12}{2\sqrt{2}}$
204			A1 for clear completion to $3\sqrt{2}$
205(a)	$9\sqrt{5}$	2	M1 for $5\sqrt{5}$ or $4\sqrt{5}$
205(b)	$2\sqrt{7}+2\sqrt{2}$	3	M1 for multiplying numerator and denominator by $\sqrt{7}+\sqrt{2}$
205			M1 for denominator $7-2$ or 5
206(a)	$4\sqrt{3}$	2	M1 for $\sqrt{108}=6\sqrt{3}$ or $\sqrt{12}=2\sqrt{3}$
206(b)	$12+4\sqrt{3}$	3	M1 for multiplying numerator and denominator by $3+\sqrt{3}$
206			M1 for denominator evaluating to 6
206(c)	$24+24\sqrt{3}$	3	M1 for $\frac{1}{2}\times(12+4\sqrt{3})\times4\sqrt{3}$
206			M1 for expanding to $24\sqrt{3}+8\sqrt{9}$
207(a)	$\frac{5}{2}+\sqrt{2}$ or $2.5+\sqrt{2}$	4	M1 for $3x\sqrt{2}-4=x\sqrt{2}+5\sqrt{2}$
207			M1 for $2x\sqrt{2}=5\sqrt{2}+4$
207			M1 for $x=\frac{5\sqrt{2}+4}{2\sqrt{2}}$

Part	Answer	Marks	Partial Marks / Guidance
207(b)	$25-17\sqrt{2}$	3	M1 for multiplying numerator and denominator by $3-2\sqrt{2}$
207			M1 for denominator expanding to $9-8=1$
207			M1 for numerator expanding to $21-14\sqrt{2}-3\sqrt{2}+4$
208(a)	30	2	M1 for $(6\times4)+(\frac{1}{2}\times3\times4)$
208(b)	450	2	M1 for their (a) $\times15$
209(a)	55	2	M1 for $\frac{1}{2}\times(8+14)\times5$
209(b)	$\frac{1}{2}\times11\times h=55$	1	M1 for correctly equating area expression to 55
209	$h=10$	1	A1 for clean algebraic isolation without errors
210(a)	36	1	
210(b)	6	2	M1 for $\sqrt{36}$
210(c)	24	1	FT $4\times$ their (b)
211(a)	34	1	
211(b)	48	1	
212(a)	55	2	M1 for $\frac{1}{2}\times(8+14)\times5$
212(b)	$\frac{1}{2}\times11\times h=55$	1	M1 for correctly equating area expression to 55
212	$h=10$	1	A1 for clean algebraic isolation without errors
213(a)	36	1	
213(b)	50	3	M2 for $8\times4+6\times3$ or $10\times3+4\times5$
213			M1 for establishing area of one correct rectangular component

Part	Answer	Marks	Partial Marks / Guidance
214(a)	Correct proof to $x = 4$	3	M1 for $2(2x-1)+x+4$, M1 for $2(x+3+x)$, A1 for explicit simplification $5x+2 = 4x+6 \implies x = 4$
214(b)	28	2	M1 for evaluating dimensions $(4+3)\times 4$
215(a)	$\frac{3}{4}$	4	M1 for Area of square $= \frac{3}{4}\times\frac{3}{4}$, M1 for Area of triangle $= \frac{1}{2}\times\frac{1}{2}\times\frac{3}{4}$, M1 for adding $\frac{9}{16}+\frac{3}{16}$
216(a)	2	3	M2 for $10x^2 = 40$ or M1 for Area is $5x\times 2x$
216(b)	32	2	M1 for $2(5x+3x)$ or $16x$ seen
217(a)	3	3	M2 for $8x = 6(x+1)$ or M1 for Area of trapezium is $6(x+1)$
217(b)	22	1	FT $2(8+\text{their } x)$
218(a)	$(2x-1)(x+2) = 18$	1	M1 for formulating the initial area product
218	$2x^2+4x-x-2 = 18$	1	M1 for correct expansion leading to given equation
218(b)	$\frac{5}{2}$	3	M2 for $(2x-5)(x+4) = 0$ or correct substitution into quadratic formula
218			M1 for identifying $2x^2+8x-5x-20 = 0$
218(c)	17	2	M1 for substituting $x = 2.5$ into $2(2x-1+x+2)$
219(a)	Correct drawing	2	M1 for plotting all 4 points correctly
219			M1 for joining points to form a closed parallelogram
219(b)	‘16‘	2	M1 for base ‘$AB = 4$‘ and perpendicular height ‘$y = 4$‘
219(c)	‘$8+4\sqrt{5}$‘	3	M1 for exact length of ‘BC‘ or ‘AD‘ using ‘$\sqrt{(7-5)^2+(5-1)^2}$‘
219			M1 for finding slant length ‘$\sqrt{20}$‘ or ‘$2\sqrt{5}$‘

Part	Answer	Marks	Partial Marks / Guidance
220(a)	'9' and '15'	3	M1 for setting up equation '$\frac{1}{2}(3k+5k)\times 8 = 96$'
220			M1 for simplifying to '$32k = 96$' or '$k = 3$'
220(b)	'$24+2\sqrt{73}$'	4	M1 for identifying the base of the right-angled side triangle is '$\frac{15-9}{2}$' or '3'
220			M1 for formulating Pythagorean theorem '$\sqrt{3^2+8^2}$'
220			M1 for exact slant side '$\sqrt{73}$'
221(a)	48.2	2	M1 for $15.4+15.4+8.7+8.7$
221(b)	110.88 or 111	2	M1 for 15.4×7.2
222(a)	46.3	2	M1 for $18.2+12.5+7.1+8.5$
222(b)	98.24 or 98.2	2	M1 for $\frac{1}{2}\times(18.2+12.5)\times 6.4$
223(a)	172.53 or 173	3	M2 for $14.2\times 9.5+\frac{1}{2}\times 14.2\times 5.3$ or M1 for either area correct
223(b)	25.87 to 25.9	2	M1 for their (a) $\times 0.15$
224(a)	10.6 or 10.64...	3	M2 for $\sqrt{15.6^2-11.4^2}$ or M1 for $15.6^2 = 11.4^2+h^2$
224(b)	60.7 or 60.6...	2	M1 for $\frac{1}{2}\times 11.4\times 10.64\ldots$
225(a)	fully correct working	2	M1 for $85.8 = \frac{1}{2}\times(x+10.8)\times 5.2$
225(b)	48	2	M1 for $22.2+10.8+7.5+7.5$
226(a)	45.3 or 45.31...	3	M2 for $\sqrt{8.5^2-(18.2-12.4)^2}$ or 6.21...
226			M1 for $h^2+(18.2-12.4)^2 = 8.5^2$
226(b)	95.1 or 95.06...	2	M1 for $0.5\times(12.4+18.2)\times 6.21\ldots$
227(a)	$(3x+2)(x+1) = 56$	M1	

Part	Answer	Marks	Partial Marks / Guidance
227	$3x^2+3x+2x+2=56$	A1	Leading to $3x^2 + 5x - 54 = 0$ with no errors seen
227(b)	3.49 or 3.490...	3	M2 for $\dfrac{-5+\sqrt{5^2-4(3)(-54)}}{2(3)}$
227			M1 for $\sqrt{5^2-4(3)(-54)}$ or $\sqrt{673}$
227(c)	41.9 or 41.90...	2	M1 for $2\times(3(3.49\ldots)+2)+2\times(2(3.49\ldots)+1.5)$
228(a)	190 or 189.5...	3	M2 for $10.5\times14.8+0.5\times10.5\times6.5$
228			M1 for 10.5×14.8 or $0.5\times10.5\times6.5$
228(b)	56.8 or 56.81...	3	M2 for $\sqrt{6.5^2+5.25^2}$ or 8.35...
228			M1 for $6.5^2+5.25^2$
228			M1 for $14.8+10.5+14.8+2\times8.35\ldots$
229(a)	642	3	M2 for $32.0\times24.0-0.5\times(12.5+18.5)\times8.4$
229			M1 for 32.0×24.0 or $0.5\times(12.5+18.5)\times8.4$
229(b)	48.8 or 48.83...	3	M2 for $\sqrt{8.4^2+3^2}$ or 8.91...
229			M1 for $\dfrac{18.5-12.5}{2}$ or 3
229(c)	28.89	2	M1 for 642×0.045
230(a)	64.48	M1	for calculating area of rectangle '12.4×5.2'
230	34.34	M1	for total area minus rectangle area '$98.82-64.48$'
230	3.4	A1	for correct substitution and solving '$\frac{1}{2}(12.4+7.8)\times h=34.34$'
230(b)	4.10 or 4.104...	M2	for calculating slanted edge '$\sqrt{3.4^2+2.3^2}$'
230	38.8 or 38.81	A2	for full perimeter '$12.4+5.2+5.2+7.8+2\times4.104$'

Part	Answer	Marks	Partial Marks / Guidance
231(a)	correct derivation	3	M1 for area of rectangle '$(2x+3)(x+4)$' M1 for area of triangle '$\frac{1}{2}(2x+3)(x)$' A1 for combining to '$2x^2+11x+12+x^2+1.5x=110$' and rearranging
231(b)	4	2	M1 for '$(x-4)(6x+49)=0$' or correct use of quadratic formula. (Ignore negative root '$x=-\frac{49}{6}$')
231(c)	40.6 or 40.60...	4	M1 for substituting $x=4$ into lengths to get rectangle sides '11' and '8' M1 for calculating half-base of triangle '5.5' M1 for slant edge '$\sqrt{5.5^2+4^2}$'
232(a)	12π	2	M1 for $\frac{30}{360}\times\pi\times 12^2$
232(b)	$24+2\pi$	3	B1 for $a=24$, B1 for $b=2$. M1 for $\frac{30}{360}\times 2\times\pi\times 12$
233(a)	4π	2	M1 for $\frac{90}{360}\times\pi\times 4^2$
233(b)	$8+2\pi$	3	M2 for $4+4+\frac{90}{360}\times 2\times\pi\times 4$ or M1 for $\frac{90}{360}\times 2\times\pi\times 4$
234(a)	$\frac{60}{360}\times 2\times\pi\times 6=2\pi$	2	M1 for $\frac{60}{360}\times 2\times\pi\times 6$
234(b)	30π	2	M1 for $\frac{360-60}{360}\times\pi\times 6^2$
235(a)	$\frac{25\pi}{2}$ or 12.5π	2	M1 for $\frac{180}{360}\times\pi\times 5^2$
235(b)	$10+5\pi$	2	M1 for $\frac{180}{360}\times 2\times\pi\times 5$
236(a)	$\frac{240}{360}\times\pi\times 3^2=6\pi$	2	M1 for $\frac{240}{360}\times\pi\times 3^2$
236(b)(i)	30 GBP	2	M1 for 6×5